

Full-bridge inverter

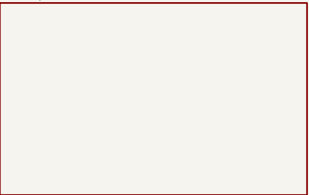
LCC-S 120 W wireless charger - root sheet: TX full bridge + sheet index



DRV_* are driver outputs. High-side VGS is referenced to SWA/SWB.
Q1-Q4 BSC070N10NS3G (BSC070N10NS3GATMA1): 100 V / 7 mOhm OptiMOS 3. VDS rating is about 4x the 24 V bus, about 2x the 45 V TVS clamp.
SWA/SWB -> sheet TX_LCC. DRV_* <- sheet TX_Driver (UCC27211 x2). PWM with dead time from STM32G474 HRTIM.
Design values: VBUS 24 V, f = 144 kHz, 120 W class. 10R / 10k gate values are starting points.
C1 220 uF/50 V bulk; C2/C3 4.7 uF/50 V X7R at each leg carry the 2 x fsw bridge ripple, about 2.4 A rms at 120 W. 50 V covers the 24 V bus and the 45 V TVS clamp.

TX side (transmitter board, ground = GND)

TX_Input

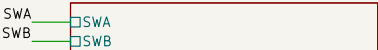


TX_Driver



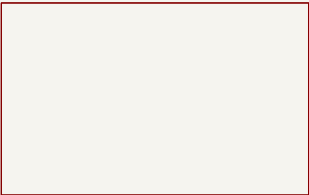
File: tx_driver.kicad_sch

TX_LCC



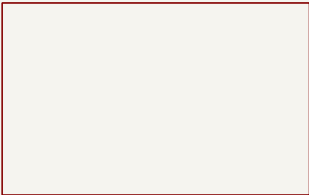
File: tx_lcc.kicad_sch

TX_AuxPower



File: tx_power.kicad_sch

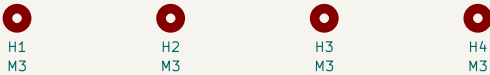
TX_MCU



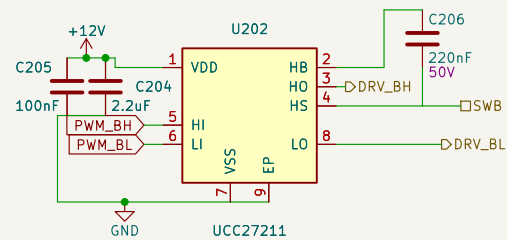
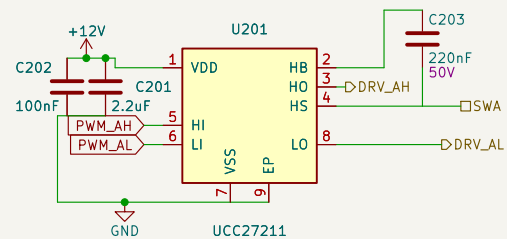
File: tx_mcu.kicad_sch

RX side (receiver board, isolated ground = GND_RX)

M3 安装孔 ×4 (NPTH, 板角各一个, PCB 上定位)



Sheet: /TX_Input/		D
File: tx_input.kicad_sch		
Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.6		Id: 2/6



U201/U202 UCC27211DRM (VSON-8): 120 V half-bridge driver with internal bootstrap diode. HI/LI are TTL inputs driven by STM32G474 HRTIM; dead time is set in HRTIM.

Bootstrap 220 nF (design example: 40 nC / 0.5 V -> 80 nF minimum). Above TI's typical 22-100 nF on purpose: a 100 nF X7R drops to about 80 nF under 12 V DC bias. VDD 2.2 uF + 100 nF, about 10x Cboot.

High-side gate is referenced to SWA/SWB through HS. Gate resistors R1-R4 and Rgs R6-R9 are on the root sheet.

Sheet: /TX_Driver/
File: tx_driver.kicad_sch

Title:

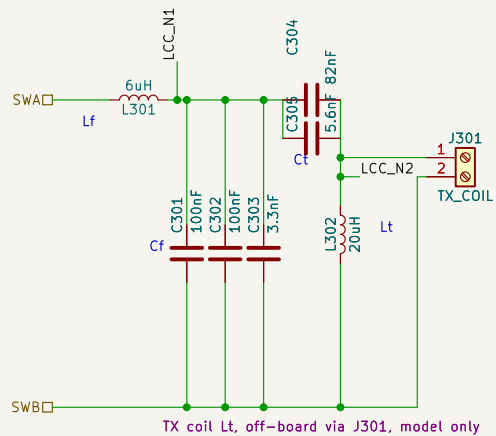
Size: A4

Date:

KiCad E.D.A. 10.0.6

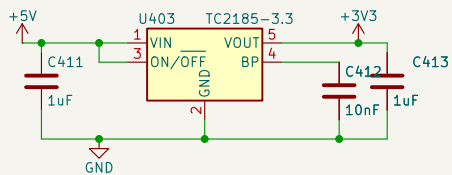
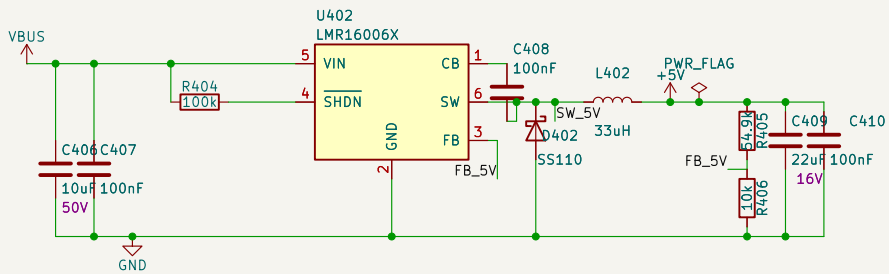
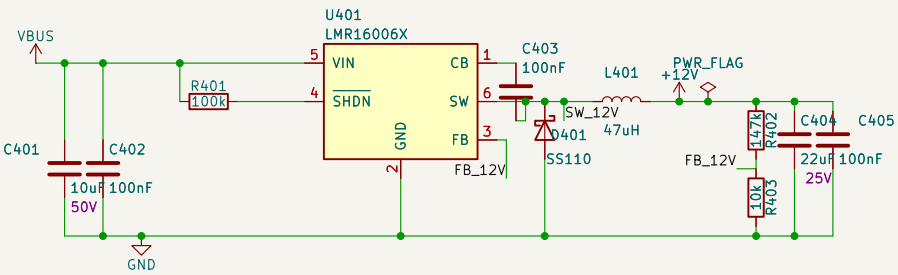
Rev:

Id: 3/6



LCC-S transmitter side, design values (see docs/design.md), $f = 144 \text{ kHz}$: $L_f = 6 \text{ uH}$, $L_t = 20 \text{ uH}$.
 $C_f = 1/(\omega^2 L_f) = 203.6 \text{ nF} \rightarrow C301 || C302 || C303 = 203.3 \text{ nF}$. $C_t = 1/(\omega^2 (L_t - L_f)) = 87.3 \text{ nF} \rightarrow C304 || C305 = 87.6 \text{ nF}$.
 Ideal-model estimate at 24 V full duty: $V_{AB1} = 21.6 \text{ Vrms}$, $L_{Lt} = V_{AB1}/(\omega L_f) = 4.0 \text{ Arms}$, $V_{Ct} = 50 \text{ Vrms}$, $V_{Lt} = 72 \text{ Vrms}$.
 Resonant caps, 250 V C0G 1812. $C_f = C301 + C302 + C303 = 203 \text{ nF}$: about 37 V rms / 7 A rms; $C_t = C304 + C305 = 87.6 \text{ nF}$: about 50 V rms / 4 A rms at 120 W. Trim to the measured L_f and L_t .
 J301 connects the off-board Litz TX coil; L302 is a model of that coil (not a board part).

Sheet: /TX_LCC/		
File: tx_lcc.kicad_sch		
Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.6		Id: 4/6



Aux power design. U401: $V_{OUT} = 0.765 \text{ V} \times (1 + 147\text{k}/10\text{k}) = 12.0 \text{ V}$ for UCC27211 VDD. U402: $0.765 \text{ V} \times (1 + 54.9\text{k}/10\text{k}) = 4.96 \text{ V}$.

U403 TC2185–3.3, 150 mA max: MCU + INA240 only. Buck first, then LDO for a quieter 3.3 V rail.

LMR16006X: 60 V, 0.6 A, 700 kHz (symbol borrowed from LMR16006YQ, same pinout). L401 47 uH / L402 33 uH give about 30 % ripple at 24 V input. SS110 catch diodes (100 V, 1 A).

VFB = 0.765 V typ (0.747–0.782 V, TI SNVSA24). Still to check: inductor saturation current, capacitor DC-bias derating.

Sheet: /TX_AuxPower/
File: tx_power.kicad_sch

Title:

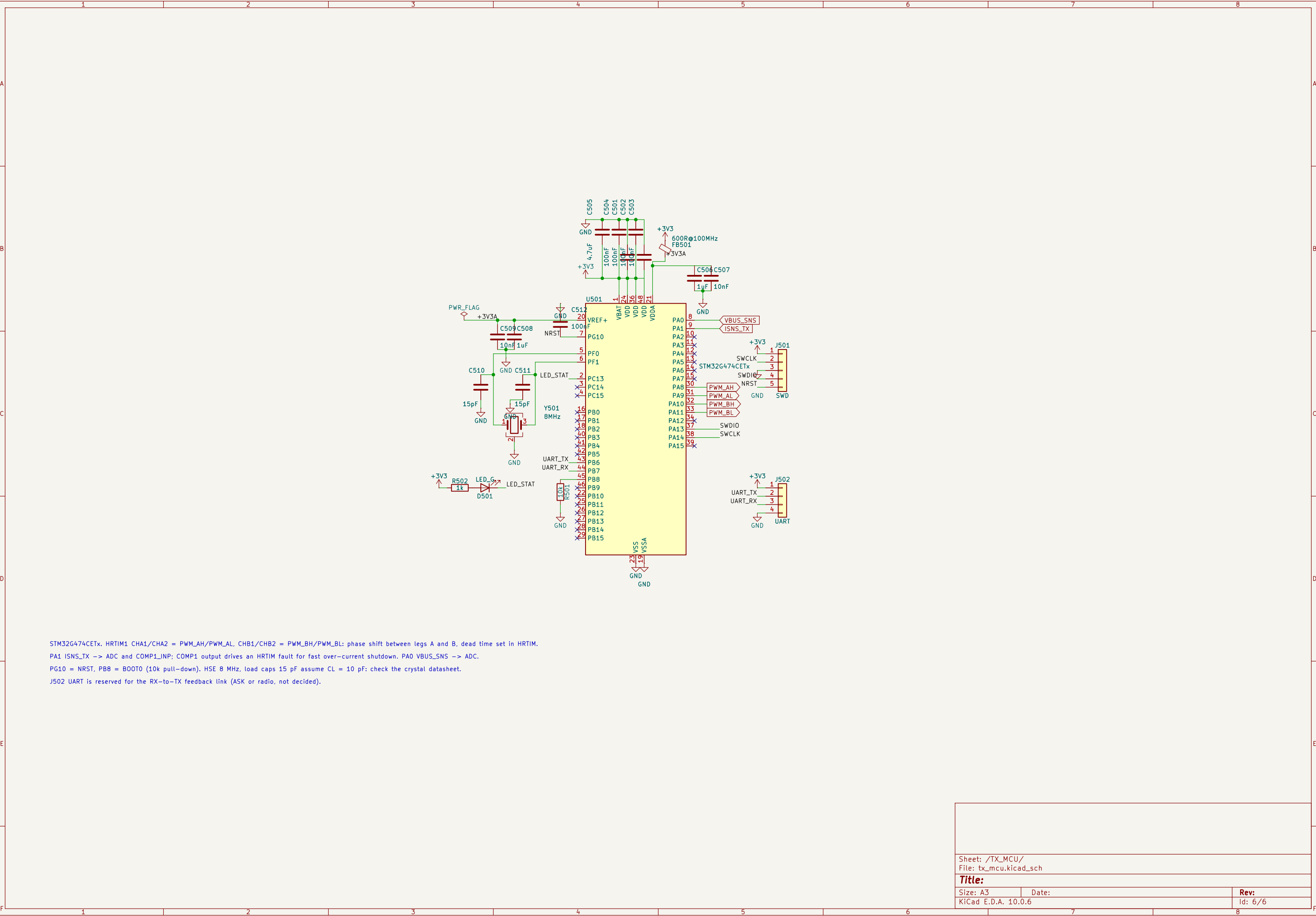
Size: A4	Date:
----------	-------

SIZE: A1	1
KiCad E.D.A. 10.0.6	

Date:

Rev:

Id: 5/6



STM32G474CETx. HRTIM1 CHA1/CHA2 = PWM_AH/PWM_AL, CHB1/CHB2 = PWM_BH/PWM_BL: phase shift between legs A and B, dead time set in HRTIM.
PA1 ISNS_TX -> ADC and COMP1_INP; COMP1 output drives an HRTIM fault for fast over-current shutdown. PA0 VBUS_SNS -> ADC.
PG10 = NRST, PB8 = BOOT0 (10k pull-down). HSE 8 MHz, load caps 15 pF assume CL = 10 pF: check the crystal datasheet.
J502 UART is reserved for the RX-to-TX feedback link (ASK or radio, not decided).